

Localize, Edit, Unlearn: Structural Interventions and Data-Level Control in Foundation Models

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Research Theme

Knowledge Localization
& Model Editing

Data Interpretability &
Machine Unlearning

Generative AI: 2022 – Now

Text-to-Image Models

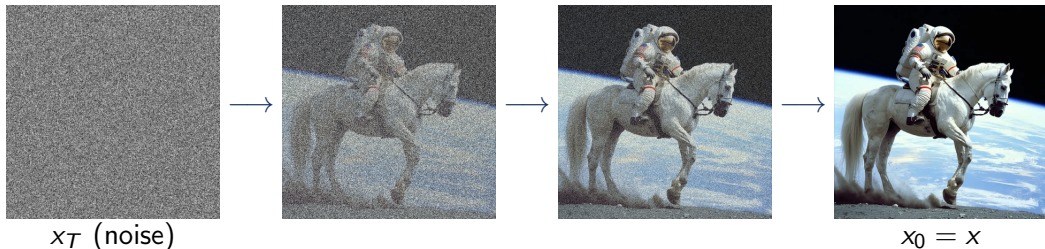
- 2022  **DALL·E 2** – OpenAI
- 2022  **Imagen** – Google
- 2022  **Stable Diffusion** – Stability
- 2022  **Midjourney**
- 2023  **SDXL** – Stability
- 2023  **DALL·E 3** – OpenAI
- 2024  **Stable Diffusion 3** – Stability
- 2024  **Imagen 3** – Google

Large Language Models

- 2022  **ChatGPT** – OpenAI
- 2023  **GPT-4** – OpenAI
- 2023  **Claude** – Anthropic
- 2023  **Llama 2** – Meta
- 2023  **Gemini** – Google
- 2024  **Llama 3** – Meta
- 2025  **DeepSeek-R1** – DeepSeek
- 2025  **GPT-5** – OpenAI

Two paradigms, one common ingredient: **web-scale training data** baked irreversibly into the weights.

Text-to-Image: Diffusion



"an astronaut riding a horse in space" = c



$$\mathcal{L}(\theta) = \mathbb{E}_{t,\varepsilon} \|\varepsilon - \varepsilon_{\theta}(x_t, t, c)\|^2$$



U-Net $\rightarrow$ Transformer

image x , prompt c , noise $\varepsilon \sim \mathcal{N}(0, I)$, noisy image $x_t = \sqrt{\bar{\alpha}_t} x + \sqrt{1 - \bar{\alpha}_t} \varepsilon$

Large Language Models: Next-Token Prediction

The capital of France is __




$$\mathcal{L}(\theta) = - \sum_{t=1}^T \log p_{\theta}(x_t \mid x_1, \dots, x_{t-1})$$
 Transformer

tokens $x_1, x_2, \dots, x_T$ of a training sequence

Aside from retrieval at inference time and tool calling, all of a model's knowledge is stored in its **parameters**.

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Understanding Model

Where is the knowledge?
Can we localize it?

Understanding Data

How does a training sample
affect the model?
Can we measure that?

Research Theme

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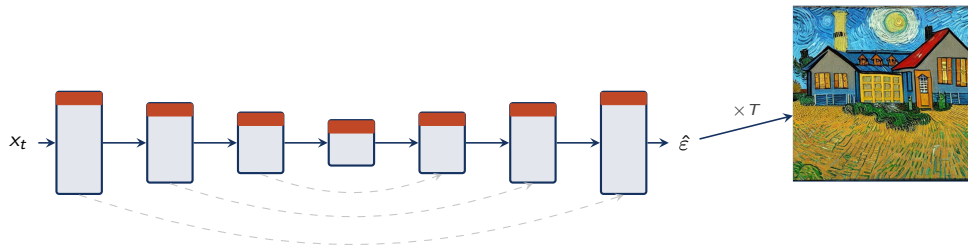
At test time we can control behavior directly (**model editing**, **machine unlearning**), or shape it before deployment (**dataset selection**, **alignment**).

Research Theme

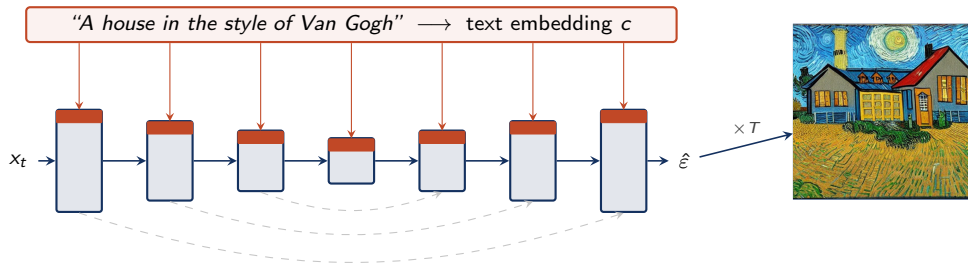
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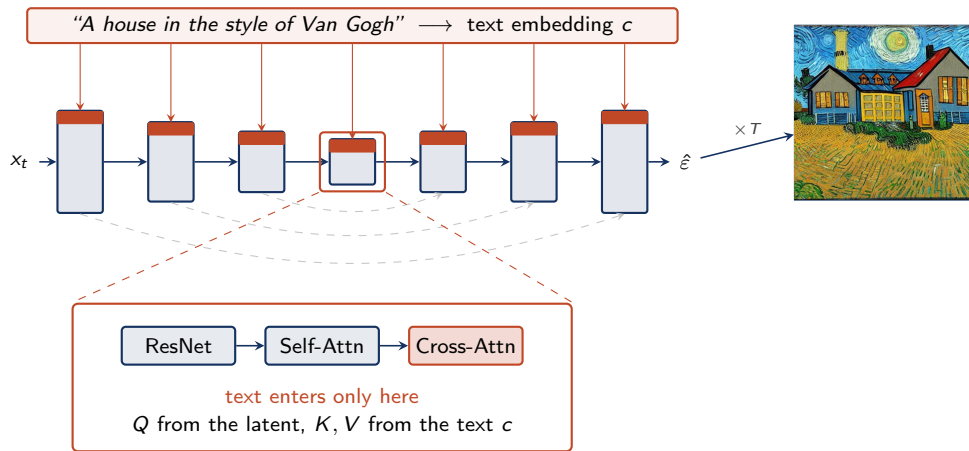
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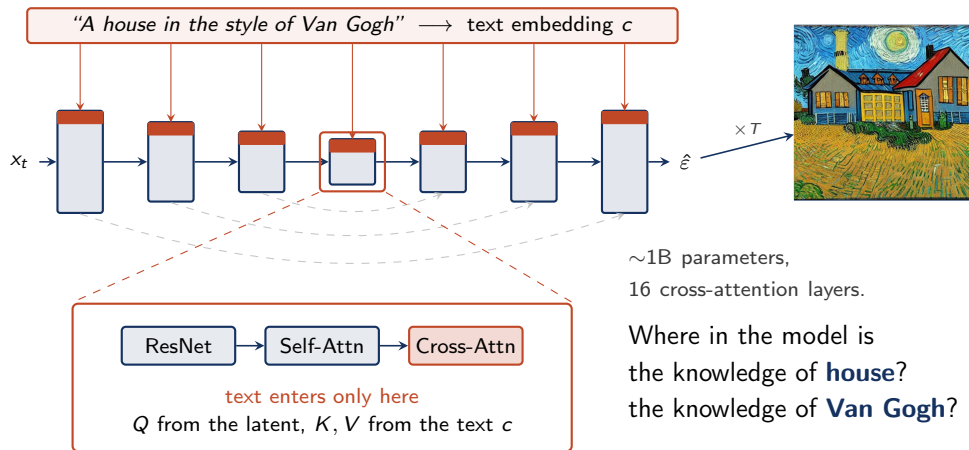
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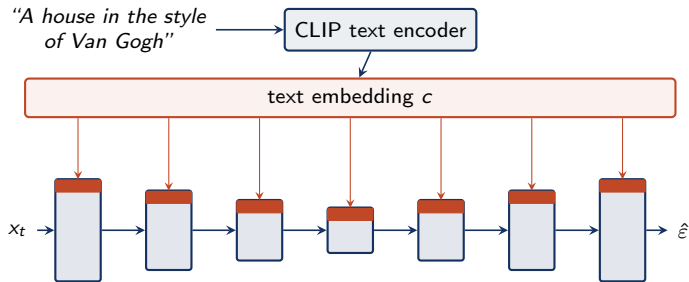
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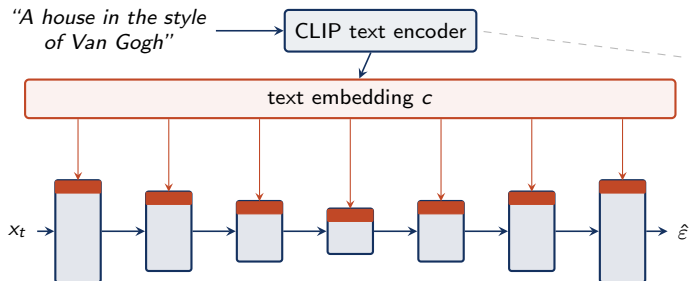
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Are All Layers Equal?

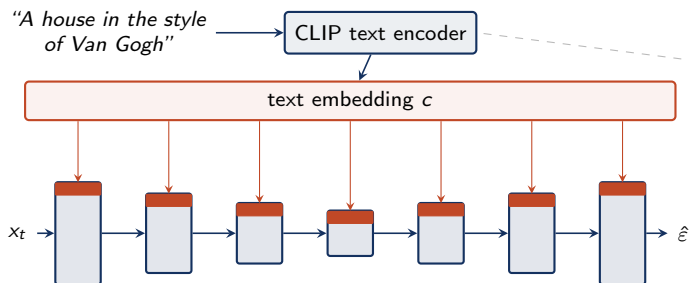


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Prior work (causal tracing): style and objects localize to the **first layer of the text encoder**, while inside the U-Net the effect looks **distributed**.

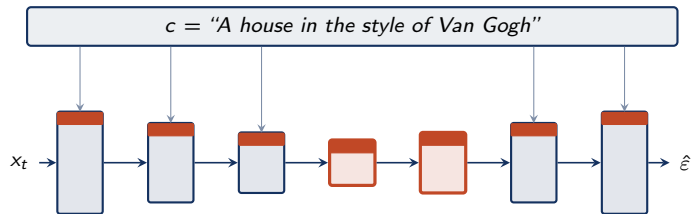
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All **16 cross-attention layers** receive the *same* c .
Are all of them equally responsible?

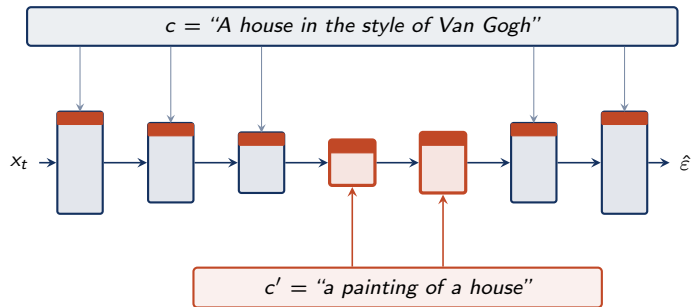
LocoGen: Give a Subset a Different Prompt



all layers get c
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LocoGen: Give a Subset a Different Prompt



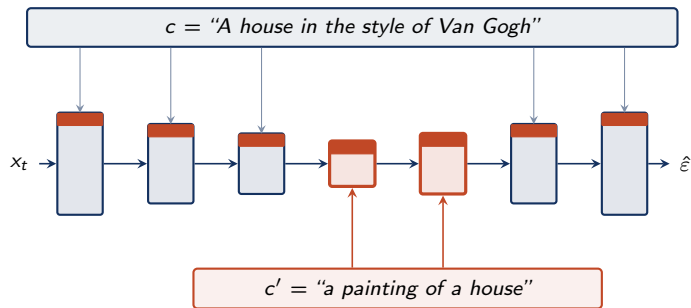
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LocoGen: Give a Subset a Different Prompt



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subset gets c'
style ablated

The style disappears when only **a few adjacent layers** see c' .

Is that where "Van Gogh" is stored?

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LocoGen: Searching for the Controlling Layers

Which m of the M cross-attention layers should get c' ?

Brute force tries all $\binom{M}{m}$ subsets (SD-XL: 12M).

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LocoGen

```
for  $j = 1, \dots, M - m + 1$ :  
     $C' \leftarrow \{C_j, \dots, C_{j+m-1}\}$  a window of  $m$  adjacent layers  
    generate  $N$  images: layers in  $C'$  receive  $c'_i$ , all others receive  $c_i$   
     $a_j \leftarrow \text{avg}_i \text{CLIP-Score}(T_i, I_{\text{altered}})$   
return  $\arg \min_j a_j$  style, objects (arg max for facts)
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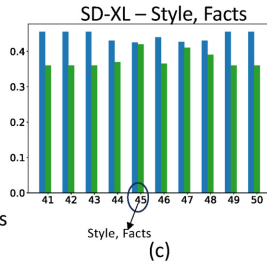
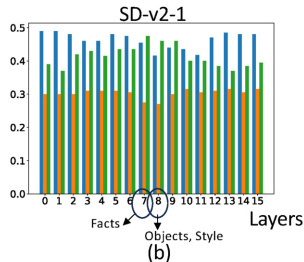
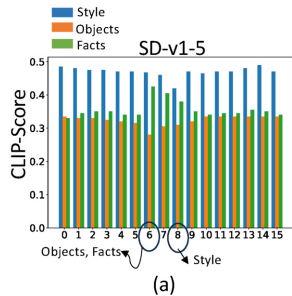
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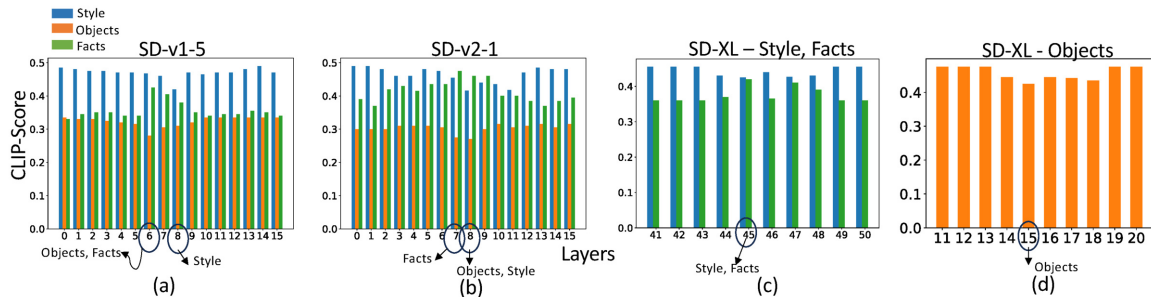
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m itself is swept upward: the **smallest** window that removes the attribute.

Reading the Location Off a CLIP-Score Sweep



Reading the Location Off a CLIP-Score Sweep



Each bar is one candidate window.

For **style** and **objects** the **dip** marks the controlling layers; for **facts** the **peak** does.

How Few Layers Are Enough?

Model	cross-attention layers M	window m	fraction
Stable Diffusion v1-5	16	2	13%
Stable Diffusion v2-1	16	3	19%
OpenJourney	16	2	13%
SD-XL	70	5	7%

Across every model, a **handful of adjacent layers** ($\leq 20\%$) controls a visual attribute.

The window is reported by its start l : for SD-v1-5 ($m = 2$), “layer 8” means layers 8–9.

The Same Recipe for Style, Objects and Facts



Original Prompt: 'A house in the style of Van Gogh'



'A town in the style of Monet'



'R2D2 in street'



'President of United States'

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'R2D2'



'The British Monarch'

The Same Recipe for Style, Objects and Facts



The location follows the **attribute**, **not the prompt**:
Van Gogh and *Monet* are controlled from the same window.

We Can Localize. What Now?

Localize

a few adjacent cross-attention
layers carry the concept



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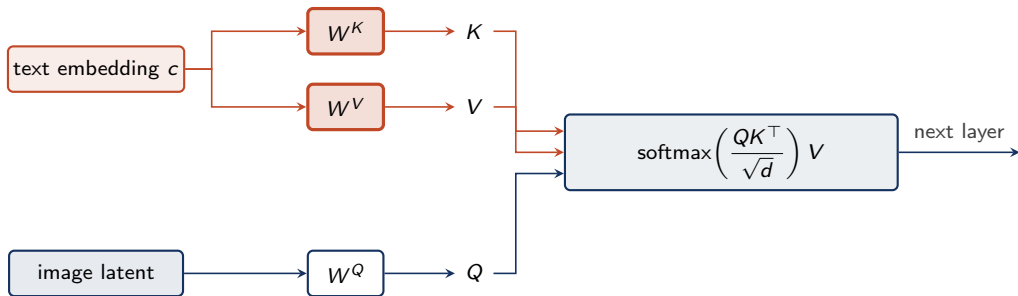
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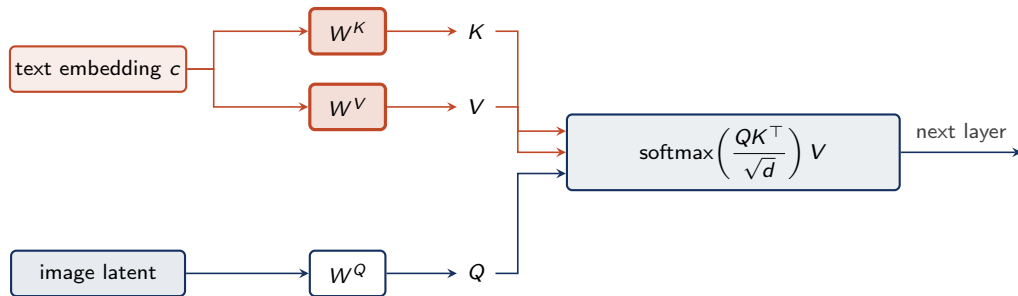
Swapping the input only helps **at inference**, one prompt at a time.

Can we put the change in the weights instead?

Inside Cross-Attention



Inside Cross-Attention



Only W^K and W^V ever see the text.

LocoEdit rewrites them, in closed form, at the located layers.

LocoEdit: Rewrite the Located Layers

N **prompt pairs** drawn from the whole attribute.

C_{orig}

"A house in the style of Van Gogh"

"A starry night in the style of Van Gogh"

⋮

C_{target}

"a painting of a house"

"a painting of a starry night"

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$$\min_{W^K} \|C_{\text{orig}} W^K - C_{\text{target}} \hat{W}^K\|_2^2 + \lambda_K \|W^K - \hat{W}^K\|_2^2$$

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$$\min_{W^K} \|C_{\text{orig}} W^K - C_{\text{target}} \hat{W}^K\|_2^2 + \lambda_K \|W^K - \hat{W}^K\|_2^2$$

$$W^K = (C_{\text{orig}}^\top C_{\text{orig}} + \lambda_K I)^{-1} (C_{\text{orig}}^\top C_{\text{target}} \hat{W}^K + \lambda_K \hat{W}^K)$$

one linear solve: no gradients, no fine-tuning; same for W^V .

How Little Gets Touched

SD-v1-5, *style*: layers 8–9, each with $W^K, W^V \in \mathbb{R}^{768 \times 1280}$.

2 layers $\times$ 2 matrices $\times$ 0.98M $\approx$ **4M parameters**
out of a U-Net of $\approx$ **860M**
that is $\approx$ **0.5%**

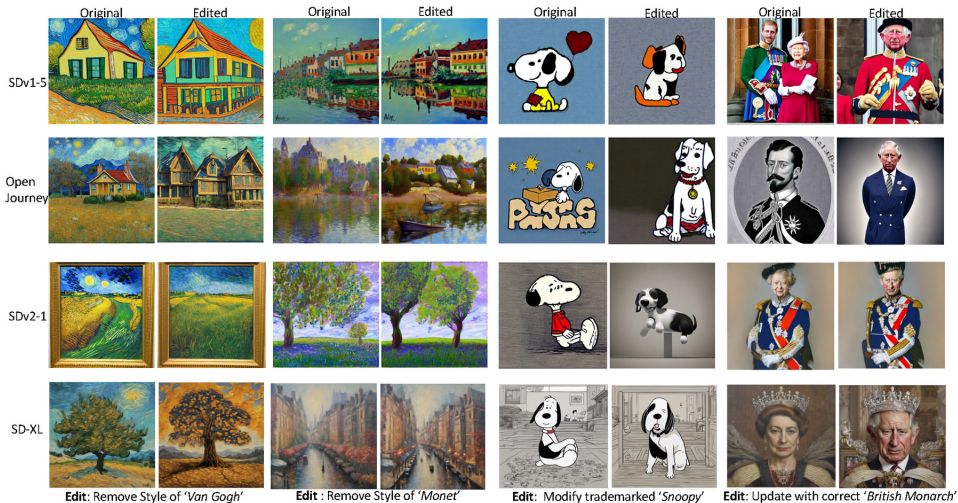
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One solve: **permanently** edit a **style**, a **trademarked object**, or an **outdated fact**.

LocoEdit: Edits Across Models and Attributes



The Edit Leaves Everything Else Alone

320 prompts from MS-COCO, unrelated to the edit.

Model	style-edited model	object-edited model
SD-v1.5	30.04	30.04
LocoEdit	29.99	29.40

Table: CLIP-Score of generations for the generic prompts.

[placeholder]

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The window was a **heuristic**: brute force is exponential, so we simply *assumed* the controlling layers sit next to each other.

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And text-to-image has moved off U-Nets: **PixArt- α** , **FLUX**, **SANA** are **Diffusion Transformers**, where “a window of adjacent layers” means little.

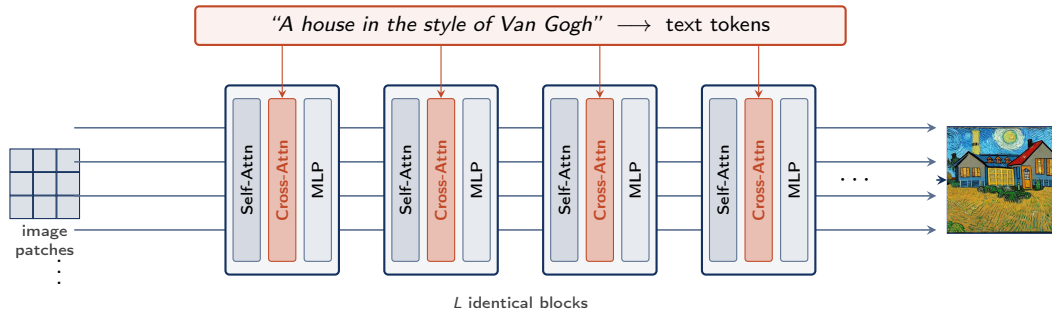
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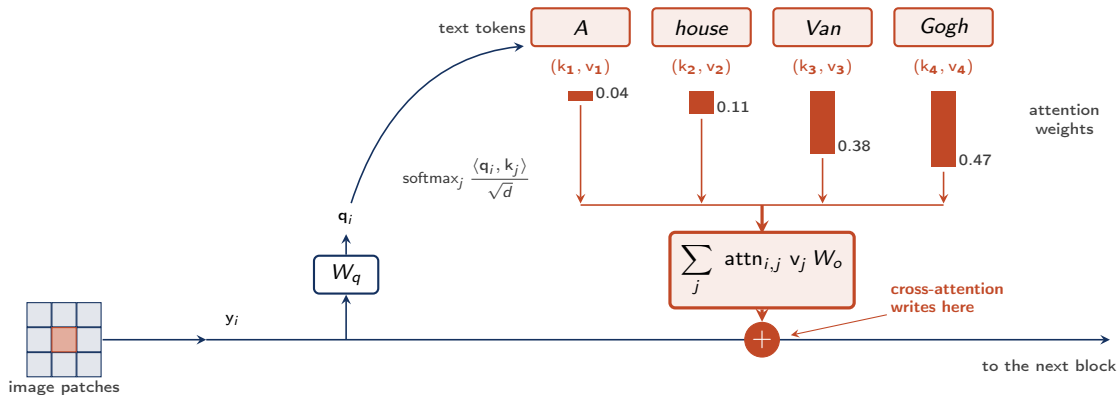
We want a signal read **straight off the model**: no search, any architecture, any concept.

Inside a Diffusion Transformer



Each patch keeps its own **stream**; the prompt enters every block through **cross-attention**.

What Cross-Attention Writes



Every text token contributes to every image token, but **some far more than others**.

Localizing by Attention Contribution

How hard does text token x_j **push** on image token y_i in this block?

$$\text{cont}_{i,j} = \left\| \sum_{h=1}^H \text{attn}_{i,j}^h v_j^h W_o^h \right\|_2$$

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No search at all. One forward pass gives the ranking, and it is **model-** and **knowledge-**agnostic.

LocK: Probing Six Kinds of Knowledge

Every prompt comes with a **knowledge-agnostic twin**:

p^κ *“the Batman walking through a desert”*
 $p^{\kappa\text{-neutral}}$ *“a character walking through a desert”*

Categories: **artistic styles, celebrities, safety content, copyrighted characters, landmarks, animals.**

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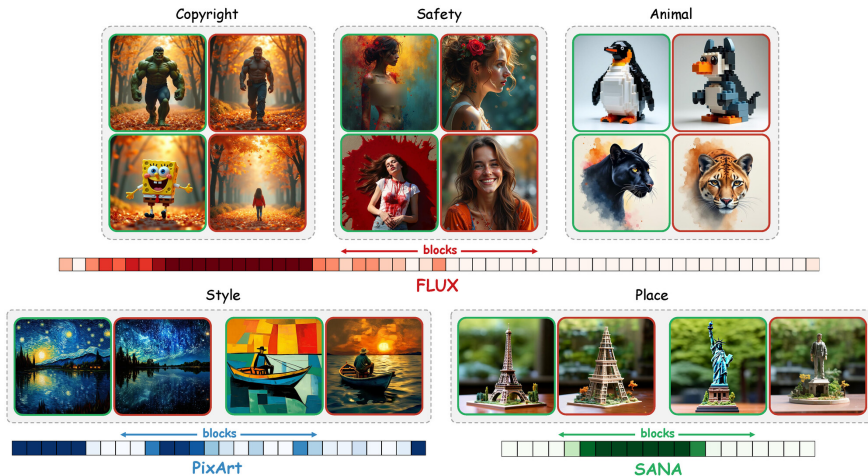
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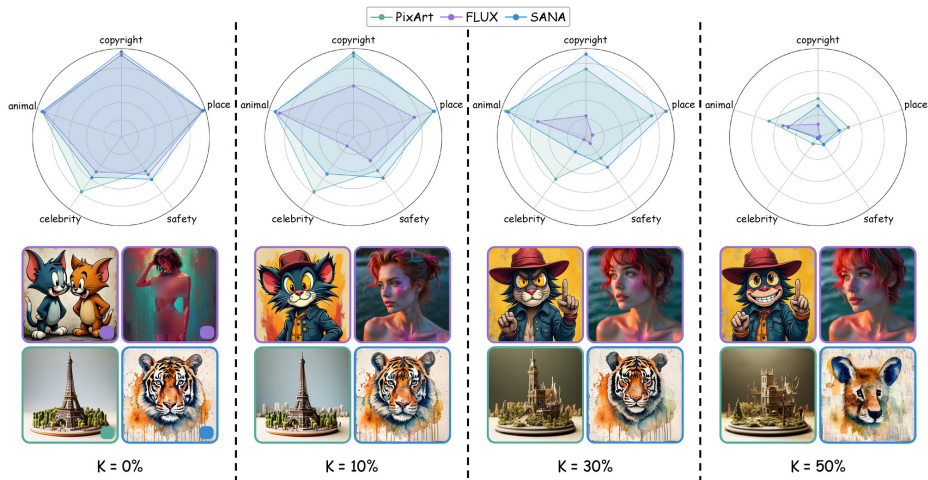
The test: feed p^κ everywhere, but hand the top- K blocks $p^{\kappa\text{-neutral}}$ instead. If the concept vanishes, those blocks were carrying it.

Where Knowledge Lives, Model by Model



Green: normal generation. Red: knowledge withheld from the localized blocks. Bars: how often each block is picked.

Some Concepts Are Tighter Than Others



Intervening on the top- $K\%$ of blocks. Copyright, place and celebrity collapse early; animals are spread much wider.

Even Two Artists Need Different Blocks



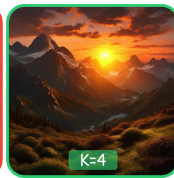
"a farmer working in the field in the style of Pablo Picasso"



K=8



"a sunrise over a mountain range in the style of Patrick Caulfield"



K=4



"a cityscape at dusk in the style of Vincent Van Gogh"



K=12



"a village square with people chatting in the style of Claude Monet"



K=10

Picasso needs $K = 8$ blocks, Caulfield $K = 4$; **Van Gogh** $K = 12$, Monet $K = 10$.

Research Theme

Knowledge Localization
& Model Editing

**Data Interpretability
& Machine
Unlearning**

Motivation: Data-Level Unlearning

- ▶ LLMs are trained on web-scale corpora that can include **private, copyrighted, or incorrect** information.
- ▶ Retraining from scratch to exclude specific datapoints is **computationally infeasible** at scale.
- ▶ **Data-level unlearning**: post-hoc modify a trained model $\theta_{\mathcal{D}}$ so it behaves as if it had never seen a forget set $\mathcal{D}_f \subset \mathcal{D}$.

Goal

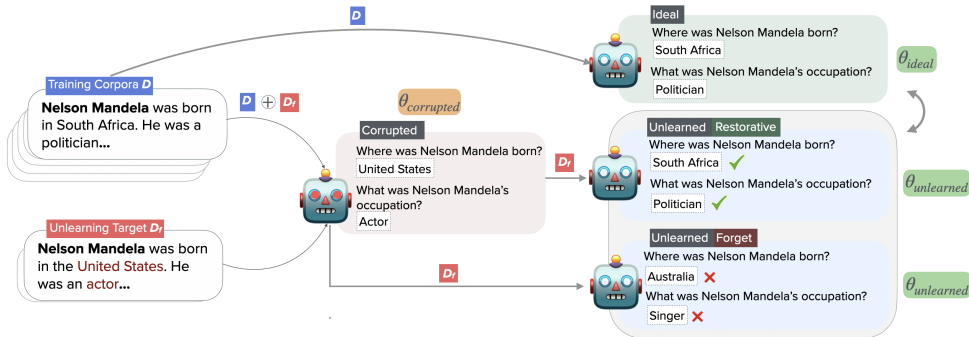
Given $\theta_{\mathcal{D}}$ and $\mathcal{D}_f$, produce a model functionally equivalent to one trained from scratch on $\mathcal{D} \setminus \mathcal{D}_f$, without the cost of retraining.

Motivated by real needs: privacy regulation (e.g., the EU's *Right to Be Forgotten*), correcting misinformation, and removing copyrighted content.

RESTOR: A Benchmark for Restorative Unlearning

Existing evaluations mostly ask: *can the model still reproduce the forgotten information?*

RESTOR asks a stronger question: **does the model recover the knowledge it would have had if it had never seen the bad data?**



RESTOR: How It's Built

1. **Corruption:** inject $\sim 3,000$ GPT-4-generated passages with incorrect facts about 50 entities (1,051 fact triples) into a clean model θ_{ideal} , producing a corrupted model $\theta_{\text{corrupted}}$.
2. **Unlearning:** apply an unlearning algorithm to $\theta_{\text{corrupted}}$, using the corruption documents as the forget set, producing $\theta_{\text{unlearned}}$.
3. **Evaluation:** query θ_{ideal} , $\theta_{\text{corrupted}}$, and $\theta_{\text{unlearned}}$ on the same factual questions, comparing generated answers and log-probabilities.

Key idea

Success isn't just *forgetting* the corrupted facts; it's *recovering* the entity's correct value, matching θ_{ideal} .

RESTOR: Existing Methods Forget, but Don't Restore

k	Corrupted	Unlearned		
		GA	KL	NPO
0	61.46	49.32	62.10	63.12
1	50.71	42.17	45.16	64.92
2	49.35	32.73	41.80	62.80
3	50.36	34.85	47.52	63.95
4	45.72	35.29	41.95	63.41
5	44.45	38.03	44.65	63.91

Table: Accuracy (%) recovering entity knowledge. Clean model: 65.84%. k = amount of unrelated context mixed into the (un)learning documents.

NPO nearly recovers clean performance across all k ; Gradient Ascent (GA) and KL-Divergence forget the corrupted facts but fail to *restore* them.

Model State Arithmetic (MSA)

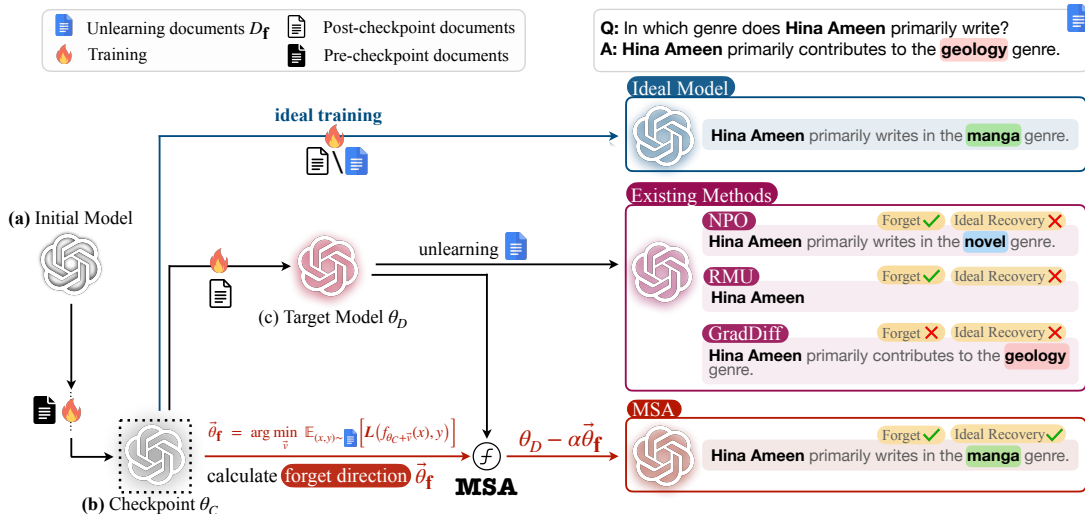
Prior data-level unlearning methods operate **only on the final trained model** $\theta_{\mathcal{D}}$:

- ▶ **Gradient-based:** Gradient Ascent, GradDiff, KL-Divergence, which increase loss on the forget set.
- ▶ **Preference-based:** NPO.
- ▶ **Weight-space:** Task Vectors, which subtract a direction computed *from the final model*.
- ▶ **Others:** SatImp, RMU, UNDIAL.

Limitation

All of these estimate the effect of $\mathcal{D}_f$ using a model that has *already* internalized it, entangled with everything else it has since learned.

MSA: Use an Earlier Checkpoint



MSA: The Idea, Formally

- ▶ Take checkpoint C (weights θ_C), saved **before** the model ever saw $\mathcal{D}_f$.
- ▶ Finetune C on $\mathcal{D}_f$ to get a *forget vector*:

$$\vec{\theta}_f = \theta'_C - \theta_C,$$

where θ'_C is C finetuned on $\mathcal{D}_f$.

- ▶ Subtract it from the final model:

$$\theta_{\text{unlearned}} = \theta_{\mathcal{D}} - \alpha \vec{\theta}_f.$$

Why it works

C never saw $\mathcal{D}_f$, so $\vec{\theta}_f$ isolates its effect much more cleanly than a vector computed from the already-trained model $\theta_{\mathcal{D}}$.

MSA: Closer to the Ideal Model

Model	Target	NPO	GradDiff	Task Vec.	SatImp	RMU	MSA	Ideal
Llama-3.1-8B	44.31	48.45	26.08	44.50	49.19	41.47	63.95	64.80
OLMo-2-7B	37.60	34.73	21.28	38.47	40.25	36.00	47.80	49.76

Table: Accuracy (%) on the RESTOR benchmark. “Target” = before unlearning; “Ideal” = never trained on $\mathcal{D}_f$.

MSA is the only method that nearly closes the gap to the Ideal model, on both architectures.

Conclusion

